# Multi-Criteria Decision Analysis (MCDA)

## 1. Introduction

Multi-Criteria Decision Analysis (MCDA) is a decision-making analysis that evaluates multiple (sometimes conflicting) criteria as part of a decision-making process. It is different to Programme Budgeting and Marginal Analysis (PBMA), because this method requires weightings and values to be assigned in order to arrive at decisions that represent best value for resources.

There is a **UK Treasury Green Book guide to MCDA**.

There are five steps of MCDA:

1. Define the context
2. Identify the options available
3. Decide the objectives and select the right criteria that represents the value
4. Measure out each of the criteria in order to discern their relative importance
5. Calculate the different values by averaging out and weighting scores

## 2. MCDA Process

- Agree decision context and identify options (e.g. programmes) to be appraised
- Identify strategic objectives/outcomes (e.g. inequalities, efficiency etc.) and criteria (e.g. cost benefit analysis, value for money).
- Agree scoring methodology (e.g. panels using a preference scale) and score each option (against each criterion).
- Agree weighting (assign weights for each of the criterion to reflect their relative importance to the decision).
- Combine the weights and scores for each option and derive an overall value
- Analyse and examine the results
- Undertake a sensitivity analysis of results

The ISPOR MCDA taskforce state that although there are many differences in the ways in which these models are used and applied, there are **several main elements of the process that are common among these methods**. The taskforce state any value measurement modelling approach entails defining the decision problem being addressed, selecting the criteria, measuring alternatives' performance, scoring alternatives and weighting criteria, aggregation, uncertainty analysis, and interpretation of results. The Table below describes the steps in value measurement for MCDA, as described by the ISPOR Taskforce.

**Table 1. The ISPOR Taskforce steps in value measurement for MCDA**

| Step | Description |
|---|---|
| Define the decision problem | Identify the decision, objects, alternatives, stakeholders and outputs required |
| Select and structure criteria | Identify the criteria relevant for evaluating alternatives |
| Measure performance | Gather data on the alternatives' performance and summarise in a "performance matrix" |
| Score the alternatives | Obtain stakeholders' preferences for changes within criteria |
| Weight criteria | Obtain stakeholders' preferences for changes between criteria |
| Calculate aggregate scores | Determine the "total value" by using the alternatives' scores on the criteria and weights of the criteria. Rank all alternatives by this "total value" |
| Deal with uncertainty | Test the level of robustness of the MCDA result by performing uncertainty analysis |
| Report and examine the findings | Explain the MCDA outputs, including the results of any uncertainty analysis to support decision making |

MCDA is distinct to PBMA, as it requires numerical scoring, valuation and weighting of criteria and outcomes.

## 3. MCDA in Practice

An example of MCDA in practice - prioritise investments in a county council setting.

Commissioners completed a submission for each service with answers to questions corresponding to ten criteria;

- Corporate Priorities
- Policy Implications
- Evidence Base: Facts, Figures & Trends
- Best Practice
- Impact on Inequalities
- Outcomes
- Targets & Performance
- Local views
- Cost effectiveness and derived benefits
- Reach

Each submission was discussed by a panel who were convened to score each service from zero to 100 against the specified criteria. The example scoring below refers to evidence of impact on inequalities.

**E. Impact on Inequalities (Ev)**

How well does the submission describe any health inequalities that affect the groups who the service is targeted at? Does the service contribute to identifying those people who suffer the most from health inequalities? Does the submission attempt to articulate the service numbers and the proportion of people from the most deprived wards/areas/groups – typically those facing greatest inequalities.

**Scoring**

**100** **Comprehensive and complete** description of relevant PH inequalities issues and the service targets & engages all appropriate groups and is meeting other unmet needs

**90** **Very good** description of relevant PH inequalities issues and the service targets & engages most appropriate groups

**80** **Good** description of relevant PH inequalities issues and the service targets and engages most appropriate groups and is meeting other

**70** **Reasonable** description of relevant PH inequalities issues, the service seeks to target and engage most appropriate groups

**60** **Average** description of relevant PH inequalities issues, the service tries to target and does engage a number of appropriate groups

**50** **Average** description of PH inequalities issues, the service has tried to target and does engage some appropriate groups

**40** Some understanding of PH inequalities issues, the service engages some appropriate groups

**30** Little understanding of PH inequalities issues, though the service does engage inequality affected groups

**20** Poor understanding of PH inequalities issues with the service inviting all users unaware of any inequality issues

**10** No evidence or understanding of PH inequalities issues with the service inviting all users unaware of any inequality issues

**0** No information offered

Each criteria had a weighting and a weighted sum method was used.

*Weighted sum score = Sum {[score against each criteria] *[criteria weight factor]} / 100*

Results from the MCDA can then be presented as a ranked list, as shown in the example below (results are for illustrative purposes only)

| | Weighting | 7 | 7 | 7 | 11 | 9 | 19 | 15 | 7 | 11 | 7 | 100 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Rank | Service | Corporate Priorities | Policy Implications | Evidence Base: Facts, Figures & Trends | Best Practice | Impact Inequalities | Outcomes | Targets & Performance | Local views | Cost effectiveness and derived benefits | Reach | Weighted sum score |
| 1 | **Weight management** | 60 | 0 | 68 | 60 | 75 | 65 | 71 | 10 | 70 | 4 | **54** |
| 2 | **Smoking cessation** | 53 | 0 | 55 | 63 | 70 | 47 | 39 | 35 | 63 | 5 | **45.3** |
| 3 | **Sexual Health** | 58 | 100 | 58 | 58 | 53 | 31 | 14 | 5 | 39 | 7 | **39.4** |
| 4 | **Dental Health Promotion** | 28 | 75 | 45 | 21 | 32 | 13 | 34 | 23 | 45 | 32 | **31.9** |
| 5 | **Community Tennis Programme** | 31 | 0 | 35 | 30 | 33 | 30 | 13 | 35 | 10 | 10 | **22.8** |

MCDA can be used in a number of ways for example, to choose one programme from a set of alternatives; to understand which criteria are most important in making a decision; to rank a set of programmes against each other.

As with any other economic method, it is only one tool in the toolbox, and is never perfect, so it is advised to sense check the results of the MCDA.

## 4. Resources

**Guidance** **The Green Book guide to MCDA**

**Guidance** **The ISPOR guide to MCDA**

## 5. Example of a published MCDA exercise

**Garau, M., Hampson, G., Devlin, N. et al.** **Applying a Multicriteria Decision Analysis (MCDA) Approach to Elicit Stakeholders' Preferences in Italy**: The Case of Obinutuzumab for Rituximab-Refractory Indolent Non-Hodgkin Lymphoma (iNHL). PharmacoEconomics Open 2, 153–163 (2018).

## 6. Contact information

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